

The Limits of Formal Methods: Notes on the Formal
Representation of Stoic Katalepsis and Hume' s Critique of
Causation

*Bıçımsel Araçların Sınırı: Stoacı Katalepsis ile Humecu Nedensellik Eleştirisinin Bıçımsel
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Abstract

This paper is a methodological note, not a historical thesis. It examines a narrow question: what can and cannot be represented when Stoic *katalepsis* —the “cognition” (Long & Sedley 1987, 40) of a kataleptic impression —and Hume’s account of causation are encoded in first-order logic (FOL)? We formalize the two positions at the level of their justification conditions and argue that a purely extensional FOL encoding can represent the *material content* of both positions but cannot represent their *explanatory* and *modal* residue: (i) Hume’s claim that a causal belief is unjustified *because* it is produced by custom has the form of an explanatory (grounding) claim, and grounding is hyperintensional —it is not a truth-condition and therefore not part of FOL model theory; (ii) the ancient definition of the kataleptic impression contains a literal modal clause —the impression is “of such a kind as could not arise from what is not” (Diogenes Laertius 7.46; Sextus Empiricus, *Adversus Mathematicos* 7.248, 7.402) —which any FOL encoding must de-modalize. The two residues are not symmetric: the Stoic residue is modal and is recovered by adding a modal operator, while the Humean residue is explanatory and requires a hyperintensional connective (grounding); the negative result is that neither is a truth-condition of the extensional encoding. We show this by a model-pair argument: two epistemologies with identical FOL structures realize different explanatory readings, and no first-order theory in the unexpanded signature can separate them. We do not claim that formalization “necessarily” fails; we claim only that this encoding, in the unexpanded signature L , at this level of granularity, is semantically —not syntactically —underdetermined with respect to the intended normative reading, and that representing the distinction requires a justification operator (justification logic: Artemov & Fitting 2019), a grounding connective (Fine 2012), or a modal operator beyond FOL. We then correct two historiographical errors that frequently accompany such formalizations and add documented channels: (i) the claim that Hume rejects the Principle of Sufficient Reason must be qualified —Hume denies that the causal maxim is rationally demonstrable (T 1.3.3; SBN 78–82) and denies any “absolute or metaphysical necessity” in causal connection (T 1.3.14.35; SBN 172); (ii) the transmission of Stoic epistemology to early modern Europe ran through documented printed texts —Sextus Empiricus (Latin *Hypotyposes*, tr. Henri Estienne, 1562; Latin *Adversus Mathematicos*, tr. Gentian Hervet, 1569; Greek *Opera*, 1621), Diogenes Laertius (Traversari’s translation completed 1433; Latin editio princeps Rome 1472; Venice 1475; Greek editio princeps Basel 1533), and Cicero’s *Academica* (editio princeps Rome 1471; the first analogy at *Lucullus* 145 = Long & Sedley 41B), supplemented by the Neostoic reception (Lipsius 1604) —not through a speculative medieval chain. The paper concludes that formal tools are useful precisely when their limits are stated, that the history of philosophy requires documentary discipline that formal notation cannot supply, and that this stance has systematic analogues in non-Western traditions of textual and logical rigor (the Buddhist *catuṣkoṭi*; the Xunzian theory of the rectification of names; the Qing *kaozheng* tradition), which we invoke as analogies, not as transmission claims.

Keywords: Stoicism; *katalepsis*; *phantasia kataleptike*; Hume; causation; Principle of Sufficient Reason; first-order logic; justification logic; grounding; transmission of ancient philosophy; *catuṣkoṭi*; *zhengming*; *kaozheng*

Özet

Bu makale tarihsel bir tez değil, ****metodolojik bir nottur****. Dar bir soruyu inceliyoruz: Stoacı ***kataleptis*** (kavrayıcı izlenimin "kavranması"; Long & Sedley 1987'nin çevirisiyle "cognition") ile Hume'un nedensellik açıklaması birinci derece mantıkta (FOL) kodlandığında ne temsil edilebilir, ne temsil edilemez? İki konumu gerekçelendirme koşulları düzeyinde biçimselleştiriyor ve tamamen ekstansiyonel bir FOL kodlamasının her iki konumun ****maddi içeriğini**** temsil edebildiğini, ancak ****açıklayıcı ve modal kalıntıyı**** temsil edemediğini gösteriyoruz: (i) Hume'un "nedensel inanç, alışkanlık tarafından üretildiği ***için*** gerekçelendirilmemiştir" iddiası açıklayıcı (grounding) biçimindedir ve grounding hiपरintansiyoneldir —truth-koşulu değildir, dolayısıyla FOL model teorisinin parçası değildir; (ii) kavrayıcı izlenimin antik tanımı lafzî (sözcüğü sözcüğüne, gerçek anlamda) bir modal kloz içerir —izlenim "olmayandan doğamayacak türden"dir (Diogenes Laertius 7.46; Sextus Empiricus, ***Adversus Mathematicos*** 7.248, 7.402) —ve her FOL kodlaması bu klozu de-modalize etmek zorundadır. Bunu bir model-çifti argümanıya gösteriyoruz: özdeş FOL yapılarına sahip iki epistemoloji farklı açıklayıcı okumalar gerçekleştirir ve genişletilmemiş imzada ('L' imzasında) hiçbir birinci derece teori bunları ayırt edemez. Formalizasyonun "zorunlu olarak" başarısız olduğunu iddia etmiyoruz; yalnızca bu kodlamanın, bu ayrıntı düzeyinde, amaçlanan normatif okumaya göre ****semantik olarak**** (sözdizimsel değil) eksik belirlenmiş olduğunu ve ayrımın temsili için FOL ötesinde bir gerekçe operatörü (gerekçe mantığı: Artemov & Fitting 2019), bir grounding bağlacı (Fine 2012) veya bir modal operatör gerektiğini söylüyoruz. Ardından bu tür biçimselleştirmelere sıklıkla eşlik eden iki tarihsel hatayı düzeltiyor ve belgelenmiş kanalları ekliyoruz: (i) Hume'un Yeter Sebep İlkesi'ni reddettiği iddiası nitelendirilmelidir —Hume, nedensel özdeşliğin rasyonel olarak gösterilemeyeceğini (T 1.3.3; SBN 78–82) ve nedensel bağlantıda hiçbir "mutlak ya da metafizik zorunluluk" olmadığını (T 1.3.14.35; SBN 172) söyler; (ii) Stoacı epistemolojinin erken modern Avrupa'ya aktarımı belgelenebilir basılı metinler üzerinden gerçekleşmiştir —Sextus Empiricus (Latince ***Hypotyposes***, çev. Henri Estienne, 1562; Latince ***Adversus Mathematicos***, çev. Gentian Hervet, 1569; Yunanca ***Opera***, 1621), Diogenes Laertius (Traversari'nin çevirisi 1433'te tamamlandı; Latince editio princeps Roma 1472; Venedik 1475; Yunanca editio princeps Basel 1533) ve Cicero'nun ***Academica***sı (editio princeps Roma 1471; yumruk benzetmesi ***Lucullus*** 145 = Long & Sedley 41B) ile Neostoacı alımlanış (Lipsius 1604) —spekülatif bir ortaçağ zinciri üzerinden değil. Makale, biçimsel araçların sınırları belirtildiğinde yararlı olduğu, felsefe tarihinin biçimsel notasyonun sağlayamadığı bir belge disiplini gerektirdiği ve bu tutumun Batı dışı geleneklerde de sistematik karşılıklarının bulunduğu (Budist ***catuṣkoṭi***; Xunzici adların düzeltilmesi teorisi; Qing ***kaozheng*** geleneği) sonucuna varır —bu karşılıklar aktarım iddiası değil, analogi olarak anılır.

****Anahtar Kelimeler:**** Stoacılık; ***kataleptis***; ***phantasia kataleptike***; Hume; nedensellik; Yeter Sebep İlkesi; birinci derece mantık; gerekçe mantığı; grounding; antik felsefenin aktarımı; catuṣkoṭi; zhengming; kaozheng

1 Introduction: Scope and Limits

The purpose of this note can be summarized in three points.

1. To encode the justification conditions of Stoic *katalepsis* and Hume's account of causation at first-order level, and to state precisely which distinction this encoding misses —not the material content of the two positions, but their explanatory and modal residue (§2).
2. To correct two historiographical errors that frequently accompany such formalizations —the unqualified claim about Hume and the Principle of Sufficient Reason (§3), and the speculative reconstruction of the transmission of Stoic epistemology (§4) —and to add the documented channels and the relevant secondary literature.
3. To avoid universal claims such as “formalization necessarily fails” and to state only singular, verifiable limits —of this encoding, at this level of granularity —together with a statement of what a richer logical language would add (§2.4).

A brief comparative note (§5) places the theme of the limits of formal schemes in a wider context: the theorization of such limits is not peculiar to the Western analytic tradition. The note ends with an explicit statement of its own limitations (§7) and a provenance table recording the source of every substantive claim.

Throughout, we observe the following discipline: no citation is included without a locatable primary text or a verifiable secondary source; where a claim rests on secondary literature alone, this is stated; and no claim of the form “there is no evidence for X” is allowed to function as “there is evidence that not-X” (see §7).

2 Formalization: What Can and Cannot Be Encoded

2.1 The Language and the Encoding

We fix a first-order signature

$$L = \{ \text{HCB}(\cdot), \text{CC}(\cdot), \text{J}(\cdot), \text{Kat}(\cdot), \text{Gr}(\cdot, \cdot), \text{Stoic}(\cdot) \}$$

with the intended readings:

- $\text{HCB}(x)$ — x is a Humean causal belief;
- $\text{CC}(x)$ — x is produced by custom;
- $\text{J}(x)$ — x is rationally justified (constrained by: $\text{J}(x) \rightarrow \neg\text{HCB}(x)$);
- $\text{Kat}(p)$ — p is a kataleptic impression;
- $\text{Gr}(p, x)$ — x is a cognition arising from the grasping of impression p ;

- $\text{Stoic}(x) \rightarrow x$ is a Stoic-relevant belief.

The Stoic justification condition: a belief is justified only if it is a cognition arising from a kataleptic impression:

$$(\text{Sto}) \quad \forall z (\text{Stoic}(z) \rightarrow \exists p (\text{Kat}(p) \wedge \text{Gr}(p, z)))$$

where $\text{Stoic}(z)$ is a predicate for “ z is a belief of the kind the Stoic theory is about”. The Humean condition: a causal belief is produced by custom:

$$(\text{Hum}) \quad \forall z (\text{HCB}(z) \rightarrow \text{CC}(z))$$

The exclusion claim under discussion: causal beliefs are not rationally justified:

$$(\text{Exc1}) \quad \forall z (\text{HCB}(z) \rightarrow \neg J(z))$$

or, formulated so that the exclusion flows *through* the mechanism of custom:

$$(\text{Exc2}) \quad \forall z (\text{CC}(z) \rightarrow \neg J(z))$$

Two observations are needed before we state the limit. First, the ancient sources define the kataleptic impression by a condition that is literally modal: the impression is one that “arises from what is, is stamped and impressed in accordance with what is, and is of such a kind as could not arise from what is not” (Diogenes Laertius 7.46 [Dorandi 2013]; Sextus Empiricus, *Adversus Mathematicos* 7.248 [Bury 1935]; 7.402 [Bury 1935]). The clause “of such a kind as could not arise from what is not” is an impossibility condition on the *source* of the impression: the veridicality of the impression is guaranteed by its causal provenance. Encoded in L , $\text{Kat}(p)$ is an unanalysed predicate: the modal clause has been de-modalized by stipulation. Second, Hume’s claim is not merely that causal beliefs are unjustified, but that they are unjustified *in virtue of being produced by custom* —the mechanism of belief formation excludes rational justification. This is an explanatory claim: the mode of production *grounds* the absence of justification (see §2.3).

2.2 What the Encoding Can Capture: The Material Content

It is important to be honest about what FOL *can* express here, because the limit we state in §2.3 is narrower than a careless reading of the literature suggests.

Let $T_1 = \{(\text{Sto}), (\text{Hum}), (\text{Exc1})\}$ and $T_2 = \{(\text{Sto}), (\text{Hum}), (\text{Exc2})\}$. Both theories are satisfiable: the finite model with domain $\{a\}$, $\text{Stoic} = \emptyset$, $\text{HCB} = \{a\}$, $\text{CC} = \{a\}$, $\text{Kat} = \text{Gr} = \emptyset$, $J = \emptyset$ satisfies both (the Stoic axiom is vacuously satisfied). Moreover T_2 entails T_1 : from (Exc2) and (Hum) , by transitivity of the conditional, $\forall z (\text{HCB}(z) \rightarrow \neg J(z))$ follows. But T_1 does not entail T_2 : in the model with domain $\{a, b\}$, $\text{HCB} = \{a\}$, $\text{CC} = \{a, b\}$, $J = \{b\}$, $\text{Stoic} = \text{Kat} = \text{Gr} = \emptyset$, the axioms of T_1 hold while (Exc2) fails at b ($\text{CC}(b) \wedge J(b)$). Hence the two formulations — “the exclusion attaches to causal beliefs directly” vs. “the exclusion flows through custom” —are extensionally distinguishable: they are different theories with different model classes. The material content of both readings is first-order expressible.

Nor is there any difficulty in representing the material co-occurrence facts. If one adds a predicate $E(x)$ for “ x is supported by adequate evidence” together with axioms relating J and E , the extension of J can be constrained as desired—but this is a *reduction*: it imposes a chosen reading of “justification” from outside the system, and the normative question (“what counts as adequate evidence, and why does custom fail to provide it?”) remains external to the formalization. This is the classical point that first-order theories constrain the extension of their predicates but not their meaning; we sharpen it below.

2.3 What the Encoding Cannot Capture: The Explanatory and Modal Residue

The limit we state is semantic, not syntactic.

Model-pair argument. Consider any model $M \models T_2$ with a domain D of size $|D| \geq 1$, with an element $a \in D$ such that $HCB(a)$, $CC(a)$, and $\neg J(a)$ hold in M . Two epistemologies are compatible with the *very same* L-structure M :

- (E1) $\neg J(a)$ holds because $CC(a)$ holds: the custom mechanism excludes rational justification; the belief’s provenance explains its lack of justification.
- (E2) $\neg J(a)$ holds “for no reason”: the fact is stipulated, or explained by something unrelated to custom; the extension of J simply happens to exclude a .

Nothing in M distinguishes (E1) from (E2): first-order satisfaction is truth-functional, and the *explanatory ordering* between facts is not a component of a first-order structure. Since every L-structure admits both readings, the class of structures realizing (E1) and the class realizing (E2) coincide—both are the class of all L-structures. Consequently no first-order theory in the unexpanded signature can separate them, and *a fortiori* no consistent extension of T_2 with purely extensional axioms can force the intended normative reading of J . The distinction is not first-order definable because it is not a *structural* property at all. This is a semantic limit: the normative-explanatory content is not the sort of thing an L-structure can realize.

Two refinements are needed.

(a) The explanatory (“because”) residue is hyperintensional. The claim “custom-production excludes justification” is naturally regimented as a grounding claim: for each causal belief z , the fact that $CC(z)$ grounds the fact that $\neg J(z)$ —in the notation of the contemporary grounding literature, $CC(z) < \neg J(z)$ (Fine 2012; Correia & Schnieder 2012). Grounding is hyperintensional: it is not definable in terms of material implication, and not even in terms of strict implication (Fine 2012). The failure is therefore not specific to FOL’s extensionality; it persists for modal extensions. This is why the draft’s original suggestion — “a doxastic or modal operator suffices” — is *necessary but not sufficient*: a modal operator can express the *necessity* of the exclusion, but not the *direction of determination* between the mechanism and the absence of justification. The literature on “because” (Schnieder 2011) and on grounding agrees that the connective is non-truth-functional.

(b) The modal residue is de-modalized by the encoding. The Stoic definition’s “could not arise from what is not” is a modal condition on the impression’s provenance; the Humean claim, read strongly, is that the custom mechanism *cannot* yield rational justification—a necessity claim about a belief-forming process. Expressing either requires an accessibility relation over possibilities (a Kripke structure)—an

expansion of the ontology, not a re-interpretation of L . It is true that modal formulas admit a standard translation into FOL over enriched structures (the “standard translation” of modal logic into first-order logic with a binary accessibility relation); but this translation changes the signature, and even then it captures *truth in a model*, not the hyperintensional “because”. The relevant contrast for our argument is between the unexpanded language L and its possible enrichments: the intended reading is not in the expressive range of L .

Process-based epistemologies. There is a structural reason why both theories resist static encoding. Both Stoic *katalepsis* and Humean custom are *belief-forming processes*: the Stoic condition evaluates the reliability of the impression’s source (the impression “could not arise from what is not”); the Humean condition describes a mechanism of association that is non-rational by constitution. Justification-by-reliable-process is exactly the subject matter of process reliabilism (Goldman 1979, 1–23; 1986): a belief is justified if it is produced by a reliable belief-forming process. A static, extensional language has no process dimension: it can record which beliefs were formed and which are justified, but not *the reliability of the process by which they were formed*—and it is precisely the process that carries the normative weight in both theories. This is the deep reason why the model-pair argument succeeds.

2.4 What a Richer Logic Adds: Justification, Grounding, Modality

The distinction that L cannot draw is representable in at least three richer frameworks. To keep the proposal testable, we give each option in the minimum formal shape in which it is expressive enough for the distinction.

(i) Justification logic. In justification logic (Artemov 2008; Artemov & Fitting 2019), formulas of the form $t:\varphi$ read “ t is a justification (evidence) for φ ”, where t ranges over explicit terms. Formally, we enrich L to $L^{\perp L}$ with two disjoint sorts of term: rational-ground terms $r_1, r_2, \dots$ and custom-description terms $c_1, c_2, \dots$, and add the rule of *application* $(s:(\varphi \rightarrow \psi) \wedge t:\varphi \Rightarrow (s \cdot t):\psi)$ with the factivity schema $t:\varphi \rightarrow \varphi$ (or, in weaker variants, without factivity). The Humean situation is then represented structurally: for a causal belief a , the theory contains $c:CC(a)$ —the custom term attests the production claim—while, at the meta-level, the theory has no theorem of the form $r:J(a)$ for any closed rational-ground term r . The two roles that the extensional encoding collapses—“what produced the belief” and “what justifies it”—are separated at the level of the term language. This is the minimal positive moral of our negative result: the distinction requires a language in which *evidence* is an explicit first-class object.

(ii) Grounding. With a partial-grounding connective $<$, the exclusion is written $CC(z) < \neg J(z)$, with the standard axioms: irreflexivity $(\forall x: \neg(x < x))$, asymmetry $(\forall x \forall y: x < y \rightarrow \neg(y < x))$ and transitivity $(\forall x \forall y \forall z: x < y \wedge y < z \rightarrow x < z)$ (Fine 2012; Correia & Schnieder 2012). The direction of determination—the feature absent from model theory—is the very content of the connective. (These axioms are constraints on the intended reading; the expressivity claim does not depend on accepting the full logic of grounding.)

(iii) Modality. A modal operator over mechanism-relative possibility expresses the necessity claim: $\Box(CC(z) \rightarrow \neg J(z))$, i.e., the custom mechanism *cannot but* produce unjustified beliefs. As noted, this captures necessity but not direction; it is the appropriate vehicle if Hume’s claim is read as a claim about the nature of the mechanism (see §3, point (c)).

We draw no conclusion about which framework is “the” correct one: the conclusion is disjunctive — *some* hyperintensional, modal, or explicit-evidence resource is required, and FOL is not such a resource at this granularity.

3 Hume and the Principle of Sufficient Reason: A Qualification Required

A widespread reading says “Hume rejects the metaphysical PSR” ; another says “Hume does not reject the PSR; he merely re-describes the question” . Neither is correct on its own. The verifiable facts are these.

(a) Hume denies that the causal maxim is rationally demonstrable. In *Treatise* 1.3.3, the maxim “whatever begins to exist must have a cause of existence” is said to be “neither intuitively nor demonstrably certain” : its contrary is conceivable, and our assent to it arises from observation and experience, i.e., from custom (T 1.3.3; SBN 78–82). The denial is therefore *epistemic*: the maxim is not rationally grounded.

(b) Hume explicitly denies absolute or metaphysical necessity in causal connection. T 1.3.14.35: “there is no absolute nor metaphysical necessity, that every beginning of existence shou’ d be attended with such an object” (SBN 172).

(c) Hume’ s positive account is psychological. T 1.3.14.20: necessity “is nothing but an internal impression of the mind, or a determination to carry our thoughts from one object to another” (SBN 164–65). The same thesis is stated in the *Enquiry*: “When we look about us towards external objects, and consider the operation of causes, we are never able, in a single instance, to discover any power or necessary connexion” (E 7.6; SBN 63). And the famous remark that “reason is nothing but a wonderful and unintelligible instinct in our souls” (T 1.3.16.9 SBN 178–9) makes the process-based character of the account explicit: belief formation is a mechanism, not an inference.

Hence: Hume does not “refute” the PSR in its causal form, and he does not leave it untouched; he removes it from the level of demonstration and relocates it at the level of custom. The reading “Hume denies the PSR” is false if it means he argued against the principle’ s *content*; the reading “Hume never questions it” is false if it means he accepted it as a rational principle. The correct formulation is: the causal maxim is, for Hume, a natural psychological regularity masquerading as a rational necessity.

Two further points. First, the term “Principle of Sufficient Reason” (*principium rationis sufficientis*) is Leibnizian (*Monadologie* §32; 1714); Hume never uses it, and retro-application requires care. Second, the literature on Hume and the causal maxim is substantial: on the maxim itself, see Garrett 1997, 24, 66–69; on the projectivist reading of causal necessity, Beebe 2006; on the *Enquiry* context, Millican 2002, 27–65. The contemporary revival of PSR debates (Della Rocca 2010; Pruss 2006) has made the question of what exactly Hume denied once more a live interpretive issue; our claim (a)–(c) is deliberately minimal and textually anchored.

4 The Transmission of Stoic Epistemology to Early Modern Europe: A Verifiable Route

4.1 Sextus Empiricus

The kataleptic impression and the criterion debate are documented most fully in Sextus Empiricus, *Adversus Mathematicos* VII: the criterion discussion at M 7.24–29, the definitional texts at M 7.248 and 7.402, and the “clear, evident, striking” (enarges, ektypos, plektike) marks at M 7.257–58 (cf. Bury 1935, 7.402–8). The printing history is now well documented (Floridi 2002; Popkin 1979, 18): the first Latin translation of the *Hypotyposes* was published in 1562 by Henri Estienne (Geneva); the Latin *Adversus Mathematicos*, together with a second edition of the Estienne translation, was published in 1569 by Gentian Hervet (Paris: Martin Juvenis; Antwerp: Plantin). The Greek editio princeps of Sextus’ *Opera* appeared only in 1621 (Geneva: Chouët). The common claim that “Hervet translated Sextus in 1562 and 1569” therefore conflates Estienne with Hervet and should be corrected.

4.2 Diogenes Laertius

The main doxographical source for Zeno’s and Chrysippus’ epistemology (the definitions of *phantasia*, *phantasia kataleptike*, *katalepsis*, *epistēmē* at *Vitae Philosophorum* 7.46–54; the criterion at 7.54 [Dorandi 2013]; cf. Bobzien 2003, 85–123) reached early modern Europe through Ambrogio Traversari’s Latin translation, completed in manuscript by 1433; the Latin editio princeps was printed in Rome 1472 (Georgius Lauer); the celebrated Venice 1475 (Nicolaus Jenson) is the second printed edition; the Greek editio princeps appeared in Basel 1533 (Froben) (Dorandi 2013, 15; ISTC id00219000/id00220000).

4.3 Cicero and the Neostoic Reception

Cicero’s *Academica* is the Latin source for Zeno’s fist analogy —open hand: impression; fingers bent: assent; fist: *katalepsis*; the other hand grasping the fist: *epistēmē*, available to the sage alone (*Lucullus* 145 [Brittain 2006]; SVF 1.66) —and for the criterion debate in Latin. The *Academica* entered print early: editio princeps Rome 1471 (Sweynheym & Pannartz), with the Aldine edition of 1523 establishing the standard text (Hunt 1998). But the decisive early modern channel for Stoic doctrine as a *system* was Neostoicism: Justus Lipsius published *De Constantia* (Antwerp: Plantin, 1584), and in 1604, both the *Manuductio ad Stoicam philosophiam* and the *Physiologia Stoicorum* (Antwerp: Plantin-Moretus). The reception of Stoic ethics through Lipsius and du Vair is well documented (Lagrée 1994; du Vair 1594); what matters for this note is that the *epistemological* transmission —the *katalepsis* doctrine —had to proceed through Sextus, Diogenes Laertius and Cicero, since the Neostoics’ interest was primarily ethical.

4.4 Hume's Own Access

For Hume's own acquaintance with these materials, the documented facts are these. The catalogue of Hume's library (Norton & Norton 1996) lists Cicero's *Opera* (Paris, 1740–42, 9 vols.); it does not list Sextus Empiricus. The 1840 sale catalogue, on which the reconstruction rests, shows no Sextus (Fosl 1998). However, the 1840 catalogue reflects books sold at auction; it is not identical to the full library at Hume's death, as books were often given away or sold earlier. The absence of Sextus is therefore negative evidence, not a proof of Hume's ignorance. But building a "Hume owned a Sextus" narrative on this gap is historiographically irresponsible. The standard scholarly account remains: Hume's access to Pyrrhonism ran through Bayle, Montaigne and others (Popkin 1952, 65–81; 1951). The historiography of "Hume the Pyrrhonian" is best grounded in: (i) Hume's direct reading of Cicero; (ii) his mediated reading of Sextus through the French sceptical tradition.

4.5 The Medieval Question and the Cartesian Comparison

Linear chains of the form "Stoicism → Aristotelian commentators → Scholasticism → Descartes → Locke" are not documented. The evidence for discontinuity is strong: only seven Latin Sextus manuscripts circulated in the medieval Latin West (Floridi 2002); the *Academica* was among the rarer Cicero texts in the Middle Ages, with Henry of Ghent the first scholastic explicitly to engage it (Schmitt 1972; 1983; SEP "Medieval Skepticism"); the term "scepticus" entered Latin usage only in the 1430s. Two qualifications are required, in the spirit of §7. First, "no continuous direct transmission of the *doctrine of katalepsis*" does not mean "no knowledge of Stoic epistemology at all": Cicero's *comprehensio* and Augustine's *Contra Academicos* provided an indirect, contested channel (Nawar 2022, 185–207). Second, the relation between Descartes's "clear and distinct ideas" and the kataleptic impression should be studied as *discrete responses to a common skeptical problem* —the criterion problem —rather than as a transmission chain; this is the position of the relevant secondary literature (Frede 1983; Nawar 2022).

Similarly, Locke's critique of innate principles in *Essay* I.3 ("No Innate Practical Principles") is directed, in its explicit form, at Herbert of Cherbury's *De Veritate* (1624): Herbert is named at I.3.15 and discussed through I.3.27 (Nidditch 77–84); Descartes is not the primary target of that chapter. And the very notion of a unified "British Empiricism" is a construction of nineteenth-century historiography (Norton 1981). The moral of this section: the history of philosophy should be written from the history of printed texts, not from speculative chains.

5 A Comparative Note: Formal Limits Across Philosophical Traditions

The theme of this note —that a formal scheme has limits that must be stated from within and beyond —is not peculiar to the Western analytic tradition. We record three documented parallels and note two

further items for completeness. Each is invoked as an analogy only: no claim of historical transmission between these traditions and the Stoic/Humean material is made or implied (cf. §7).

(a) The *catuṣkoṭi*. The Buddhist four-cornered negation schema — P , $\neg P$, *both*, *neither*—was theorized by Nāgārjuna as an exhaustive decomposition of assertoric space, and it has been shown that classical two-valued semantics cannot express the fourth corner as a distinct position; the schema is naturally formalized in paraconsistent logics (Priest 2010, 24–54; 2018). One asymmetry must be stated honestly: the *catuṣkoṭi* is a case of *expanding* a formal scheme —paraconsistency extends the logic— whereas our §2 result is a case of a scheme’s *representational limit*. The parallel is therefore not in the kind of limit but in the practice: the *catuṣkoṭi* shows that theorizing about what a scheme can and cannot say—including the scheme’s own boundary—is a distinguished non-Western philosophical activity. Whether Nāgārjuna himself intended the *catuṣkoṭi* as a meta-logical reflection on the limits of assertoric space, or whether this is a modern reconstruction (Priest 2018, ch. 1; Tillemans 1999), is itself a contested interpretive question. The claim is not that Stoic or Humean materials require paraconsistency; whatever Nāgārjuna’s intentions, the *catuṣkoṭi* schema itself—as an exhaustive decomposition of assertoric space—encodes a boundary-theorizing practice, and so the question “what cannot be represented here?” is not a modern Western invention.

(b) The Xunzian rectification of names (*zhengming*). Xunzi’s chapter “On the Rectification of Names” (Xunzi 22) addresses the criterion problem for language: names are correct if fixed by shared sensory experience (the *tianguan*, “heavenly faculties”, common to all humans—Xunzi 22.5) and by convention (“names have no intrinsic appropriateness; they are fixed by agreement”—Xunzi 22.9; Hansen 1992, ch. 9; Hansen 1983). The structural parallel with the Stoic criterion debate is striking and has been noted in the comparative literature (Rošker, SEP “Epistemology in Chinese Philosophy”; Kjellberg 1996): both theories face the question of what fixes the *standard* by which representations are judged correct. The differences are equally instructive. At the level of the *kind of standard*, the Stoic criterion is source-based (the impression’s provenance) while the Xunzian criterion is intersubjective-conventional. But the deeper difference is in the *functional position* each doctrine occupies within its system. The Stoic *katalepsis* is an epistemological doctrine about individual cognitive accuracy: it serves to secure truth for a single cognitive agent, and its failure mode is epistemic error. The Xunzian *zhengming*, by contrast, is an ethical-political doctrine about the ordering of social life: its purpose is to ensure that correct naming yields correct action and social coordination (cf. Xunzi 22.9; the canonical formula “if names are not correct, language will not be in accordance with truth” is Analects 13.3—the Xunzian claim is a development in the same spirit rather than a verbatim citation), and its failure mode is social disorder. The structural parallel at the level of “what fixes the standard?” thus masks a difference in the *domain* and the *end* to which the standard is put: the analogy should not be read as implying that the two doctrines occupy the same functional position in their respective systems. A first-order encoding of either faces the same §2.3 problem: the criterion’s *normative force*—in virtue of what the standard is the standard—is not a truth-condition.

(c) *Kaozheng* and documentary discipline. The Qing-dynasty *kaozheng* (考據, “evidential research”) movement—exemplified by Dai Zhen (*Mengzi ziyi shuzheng*, Preface; in *Dai Zhen wenji*, Zhonghua Shuju, 1980; cf. Elman 1984, Introduction and ch. 5; 2nd ed. 2001)—made textual verification, philological evidence and the rejection of unfounded commentary the methodological core of scholarship. Our §4 and §7 defend precisely this discipline for the history of philosophy: the claim that documentary

discipline is a method, not an ornament, has a systematic analogue in a non-Western tradition that is still under-cited in analytic historiography. The parallel is in the *methodological attitude* —evidence over speculation —not in the object of inquiry, which differs. Again: analogy, not influence.

We also note, for completeness, the Mohist tradition's treatment of the name–reality (名/實) distinction and the argumentation chapter “Lesser Selection” (Xiaoqu) of the *Mozi* (Graham 1978; 1989), and the White Horse paradox of Gongsun Long, which turns on the gap between a predicate's linguistic ontology and its extension —a theme in the vicinity of, though not identical with, the model-theoretic underdetermination discussed in §2.1 (Hansen 1983, ch. 5). Finally, Wang Yangming's doctrine of *liangzhi* (“innate knowing”; *Chuanxilu* 2.2; Van Norden, SEP “Wang Yangming”) treats the cognitive standard as an inner, quasi-evidential faculty rather than an external impression or convention —structurally closer to the Stoic *katalepsis* than to the Xunzian convention, since both posit a source-based standard rather than an intersubjective one, yet differing in that Wang Yangming's doctrine is embedded in a moral-spiritual project (the unity of knowledge and action) rather than primarily in an epistemological criterion debate. [Provenance note: the chapter and passage numbers above follow the standard editions cited in the bibliography; where translations differ (Watson; Knoblock), we rely on the philological consensus reported in Hansen 1992 and the Stanford Encyclopedia entries cited. The claim that the Stoic/Xunzian parallel has been noted in the comparative literature is supported by Kjellberg 1996 (Sextus/Zhuangzi/Xunzi on skepticism) and by the SEP entry “Epistemology in Chinese Philosophy” (Rošker); both are cited as reporting a parallel, not as establishing a historical connection.]

6 Conclusion

This note has made three contributions.

1. Technical limit. A purely extensional first-order encoding of the justification conditions of Stoic *katalepsis* and Humean custom can represent the material content of both positions —including the extensionally distinguishable formulations (Exc1) and (Exc2) —but cannot represent their explanatory and modal residue: the “because” of the Humean exclusion (hyperintensional; grounding), the “could not arise from what is not” of the Stoic definition (modal), and the reliability of belief-forming processes. The model-pair argument shows the limit is semantic: the intended reading is not a structural property of L-models. A justification operator (Artemov & Fitting 2019), a grounding connective (Fine 2012), or a modal operator is required. This is a singular encoding limit, not a universal failure claim.
2. Interpretive limit. Hume's relation to the PSR is neither simple rejection nor silence: he relocates the causal maxim from the demonstrative level to the level of custom (T 1.3.3; SBN 78–82; T 1.3.14.20; SBN 164–65; T 1.3.14.35; SBN 172). “Hume never questions the PSR” and “Hume refutes the PSR” are both false; the accurate formulation is a *demotion*, not a refutation. A significant interpretive counter-thesis —defended notably by Della Rocca (2010) and, in a different register, by Pruss (2006) —argues that this “demotion” is a polite fiction: by reducing the PSR to a

“natural psychological regularity,” Hume effectively *evaporates* its normative status, rendering the distinction between “demotion” and “rejection” functionally moot. We maintain the demotion formulation as the more textually anchored reading, since Hume explicitly preserves the causal maxim as a psychological principle rather than arguing against its content; but the evaporation reading remains a strong interpretive alternative, and our claim should be read as a *preference* supported by the textual evidence, not as a settled result.

3. Historical limit. The transmission of Stoic epistemology to early modern Europe runs through documented printed texts: Sextus (Latin 1562 Estienne; 1569 Hervet; Greek 1621), Diogenes Laertius (Traversari 1433; Latin editio princeps Rome 1472; Venice 1475; Greek Basel 1533), Cicero’s *Academica* (in *Opera omnia*, Rome 1471 [Sweynheym & Pannartz]; Aldine edition 1523; the first analogy at *Lucullus* 145 [Brittain 2006]), and the Neostoic reception (Lipsius 1604). Speculative medieval chains are not documented; indirect medieval knowledge via Cicero and Augustine is. Hume’s own access ran through Cicero directly and Sextus through Bayle (Popkin 1952, 65–81).

Formal tools are useful precisely when their limits are stated; the history of philosophy requires a documentary discipline that formal notation cannot supply; and both theses have systematic, independently developed analogues in the Buddhist, Xunzian and kaozheng traditions (§5).

7 Limits of This Note

In the same spirit, the limits of this note are stated explicitly.

1. We make no claim of *in-principle* inexpressibility for any language richer than L: the negative result is about this signature, at this granularity, under extensional semantics.
2. We do not claim that justification logic, grounding, or modal logic “solves” the philosophical problems; we claim only that they make the relevant distinction representable.
3. The comparative section (§5) is analogical; none of its claims depends on a historical connection between the traditions compared, and none asserts one.
4. All historical claims rest on the printed-text history summarized in §4; where the secondary literature is the sole support (e.g., the medieval manuscript census), this is stated, and the sources are given.
5. “No evidence of continuous direct transmission” (§4.5) is a negative-evidence claim; it is recorded as such and is not asserted to be a proof of non-transmission.
6. Quotations follow the editions listed under “Citations and Editions” ; paragraph numbers for Hume follow the Norton & Norton (2000) and Beauchamp (1999) editions, with Selby-Bigge/Nidditch (SBN) pagination given in every case.
7. The formal encoding in §2.1 is itself an interpretive choice. Different readings of the Stoic and Humean sources might yield different signatures or axioms; our result is relative to the encoding chosen, and the choice of encoding is a hermeneutic decision that precedes the formal argument.

8. The richer frameworks of §2.4 —grounding, justification logic, process reliabilism —are invoked as *contemporary regimentations* of the relevant claims; no claim is made that Hume or the Stoics held these theories, and their vocabulary is applied in a deliberately anachronistic, analytic sense.

8 Appendix: Definitions

Term	Definition	Source
<i>katalepsis</i>	Stoic “cognition” : the grasping of a kataleptic impression; the criterion of truth for the Stoics	DL 7.46, 7.54 [Dorandi 2013]; translation “cognition” : Long & Sedley 1987, 40; SEP “Stoicism” §3.7
<i>phantasia kataleptike</i>	Kataleptic impression: “arising from what is, stamped and impressed in accordance with what is, of such a kind as could not arise from what is not”	DL 7.46 [Dorandi 2013]; M 7.248 [Bury 1935]; 7.402 [Bury 1935]
Criterion problem	The question what standard fixes which impressions are true; the regress of “criterion of the criterion”	Sextus M 7.24–29; PH I.164–169 (169 = diallelos); M 7.440–446
Assent (<i>sunkatathesis</i>)	The Stoic act of approving an impression; the middle stage between impression and cognition	DL 7.46–54; Cicero, <i>Acad.</i> 2.37–39, 2.66–67
PSR	<i>Principium rationis sufficientis</i> : Leibnizian term; every fact has a sufficient reason	Leibniz, <i>Monadologie</i> §32 (1714)
FOL	First-order logic: extensional, truth-functional model-theoretic semantics	Standard; §2
Justification logic	Logic with formulas $t:\varphi$, “ t is a justification for φ ”, for explicit terms t	Artemov & Fitting 2019
Grounding	Hyperintensional explanatory relation: $\varphi < \psi$, “ φ grounds ψ ”	Fine 2012; Correia & Schnieder 2012
Custom	Humean habit: determination of the mind to pass from one object to its usual attendant	T 1.3.14.20; E 5
<i>catuṣkoṭi</i>	Buddhist four-cornered negation: P , $\neg P$, both, neither	Priest 2010

Term	Definition	Source
<i>zhengming</i> (正名)	Xunzian rectification of names: fixing correct usage by sensory agreement and convention	Xunzi 22.5, 22.9; Hansen 1992
<i>kaozheng</i> (考據)	Qing evidential scholarship: philological verification as method	Elman 1984, Introduction and ch. 5

Citations and Editions

- Hume, *Treatise* (T): Norton & Norton (eds.), Oxford University Press, 2000; paragraphs cited as T 1.3.3 etc.; Selby-Bigge/Nidditch (SBN) pagination in parentheses.
- Hume, *Enquiry* (E): Beauchamp (ed.), Oxford University Press, 1999; paragraphs cited as E 7.6 etc.; SBN pagination in parentheses.
- Sextus Empiricus: *Primary source editions*: Latin translations —*Hypotyposes*, tr. Hervet, publ. Estienne, 1562; *Adversus Mathematicos*, tr. Hervet, 1569; Greek *Opera*, Chouët, 1621. *Modern translation used*: Bury (tr.), Loeb Classical Library, 1933–49.
- Diogenes Laertius: Hicks (tr.), Loeb Classical Library, 1925; Dorandi (ed.), Cambridge University Press, 2013.
- Cicero, *Academica*: Brittain (tr.), Hackett, 2006; *Lucullus* paragraph numbers (Plasberg).
- Dai Zhen: *Mengzi ziyi shuzheng* (孟子字義疏證), in *Dai Zhen wenji* (戴震文集), Beijing: Zhonghua Shuju, 1980.
- von Arnim, H. (ed.). *Stoicorum Veterum Fragmenta* (SVF), vol. 1. Leipzig: Teubner, 1905.

Provenance Table

§	Claim	Source
Abstract	katalepsis = “cognition” (Long & Sedley 1987, 40)	Long & Sedley 1987, 40; cf. SEP “Stoicism” §3.7 [P-01]
§2.1	Kataleptic definition with modal clause	DL 7.46 [Dorandi 2013]; M 7.248 [Bury 1935]; 7.402 [Bury 1935] [P-03]
§2.2	$T_1 \not\models T_2$, model exhibited	Formal argument, this note (model: {a,b}; HCB={a}; CC={a,b}; J={b}) [P-04]
§2.3	Grounding hyperintensional	Fine 2012; Correia & Schnieder 2012 [P-05]

§	Claim	Source
§2.3	Process reliabilism	Goldman 1979, 1–23; 1986 [P-05]
§2.4	Justification logic	Artemov 2008; Artemov & Fitting 2019 [P-06]
§3	T 1.3.3; SBN 78–82	Hume, T (Norton & Norton 2000); SBN [P-07]
§3	T 1.3.14.20; SBN 164–65	Hume, T [P-07]
§3	T 1.3.14.35; SBN 172	Hume, T [P-07]
§3	E 7.6; SBN 63	Hume, E (Beauchamp 1999) [P-07]
§3	T 1.3.16.9; SBN 178–9	Hume, T [P-07]
§3	PSR literature	Garrett 1997; Beebe 2006; Millican 2002; Della Rocca 2010; Pruss 2006 [P-07]
§4.1	1562 Estienne; 1569 Hervet; Greek 1621	Floridi 2002; Popkin 1979, 18 [P-08]
§4.1	M 7.24–29; M 7.257–58	Sextus, M (Bury, Loeb) [P-08]
§4.2	Traversari 1433; Rome 1472; Venice 1475; Basel 1533	Dorandi 2013, 15; ISTC id00219000/id00220000 [P-09]
§4.3	Fist analogy at Lucullus 145 [Brittain 2006]	Cicero, Acad. (Brittain 2006); SVF 1.66 [P-10]
Abstract	Fist analogy = Long & Sedley 41B	Long & Sedley 1987, section 41B (= Cicero, <i>Lucullus</i> 145) [P-01]
§4.3	Lipsius 1584; 1604	SEP “Justus Lipsius” ; Lagrée 1994 [P-10]
§4.4	Hume’ s library: Cicero yes, Sextus no	Norton & Norton 1996; Fosl 1998 [P-11]
§4.4	Bayle-mediated access	Popkin 1952, 65–81; 1951 [P-11]
§4.5	Medieval discontinuity; nuances	Floridi 2002; Schmitt 1972; SEP “Medieval Skepticism” ; Nawar 2022 [P-12]
§4.5	Locke–Herbert I.3.15–27	Locke, Essay (Nidditch 1975), 77–84 [P-12]
§4.5	British Empiricism myth	Norton 1981 [P-12]
§5	Catuskoṭi formalized	Priest 2010; 2018 [P-13]
§5	Xunzi 22.5; 22.9	Xunzi (ctext; Knoblock tr.); Hansen 1992, ch. 9 [P-13]
§5	Kaozheng	Elman 1984, Introduction and ch. 5; 2nd ed. 2001 [P-13]
§5	Mohist/Gongsun Long	Graham 1978; 1989; Hansen 1983, 140–51 [P-13]
§7	Limits stated	This note [P-15]

SC-ID mapping (CHK-02). Every mandatory SC-ID corresponds to at least one section covered by the table: SC-001–004 → Abstract/Özet/Keywords; SC-005 → §1; SC-006–012 → §2.1–2.4; SC-013–014 → §3; SC-015–017 → §4; SC-018 → §5; SC-019 → §6; SC-020–022 → Provenance/Citations/References;

SC-023 → Appendix: Definitions; SC-024–030 → §4.4, §4.5, Appendix, §7 (optional items are covered where adopted).

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